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United States Patent	12396729
Kind Code	B2
Date of Patent	August 26, 2025
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Fastener applicator with interlock

Abstract

The present disclosure relates to a fastener applicator including a handle, a first trigger coupled rotatably with respect to the handle, and a second trigger coupled rotatably with respect to the handle. An interlock is disposed with the second trigger and comprises a cam follower engaged in a cam channel that selectively prevents rotation of the second trigger relative to the first trigger depending on a location of the cam follower within the cam channel. The location of the cam follower in the cam channel is set by a position of the first trigger. Methods of operating the various embodiments disclosed are also provided.

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Appl. No.:	18/158866
Filed:	January 24, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230157692 A1	May. 25, 2023

Related U.S. Application Data

continuation parent-doc US 16599779 20191011 US 11564689 child-doc US 18158866
division parent-doc US 15037963 US 10485545 20191116 WO PCT/US2014/066438 20141119
child-doc US 16599779
us-provisional-application US 61906290 20131119

Publication Classification

Int. Cl.: **A61B17/10** (20060101); **A61B17/064** (20060101); **A61B17/072** (20060101); **A61B17/08** (20060101); **A61B17/122** (20060101); **A61B17/128** (20060101); A61B17/00 (20060101)

U.S. Cl.:

CPC **A61B17/10** (20130101); **A61B17/0643** (20130101); **A61B17/072** (20130101); **A61B17/083** (20130101); **A61B17/122** (20130101); **A61B17/1285** (20130101); A61B2017/00243 (20130101)

Field of Classification Search

CPC: A61B (2017/2925); A61B (2017/2913); A61B (2017/2919); A61B (2017/2924); A61B (2017/2912); A61B (2017/2922); A61B (17/2909); A61B (17/0643); A61B (17/072); A61B (17/10); A61B (17/083); A61B (17/122); A61B (17/068); A61B (17/07207); A61B (17/29)

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8056787	12/2010	Boudreaux	N/A	N/A
8066168	12/2010	Vidal	N/A	N/A
8066721	12/2010	Kortenbach	N/A	N/A
8070033	12/2010	Milliman	N/A	N/A
8070035	12/2010	Holsten	N/A	N/A
8070038	12/2010	Kostrzewski	N/A	N/A
8074861	12/2010	Ehrenfels	N/A	N/A
8074862	12/2010	Shah	N/A	N/A
8080020	12/2010	Kortenbach	N/A	N/A
8080032	12/2010	Van Der Burg	N/A	N/A
8083118	12/2010	Milliman	N/A	N/A
8083119	12/2010	Prommersberger	N/A	N/A
8091753	12/2011	Viola	N/A	N/A
8091756	12/2011	Frank	N/A	N/A
8096459	12/2011	Ortiz	N/A	N/A
8097015	12/2011	Devellian	N/A	N/A
8113405	12/2011	Milliman	N/A	N/A
8113406	12/2011	Holsten	N/A	N/A
8113407	12/2011	Holsten	N/A	N/A
8113408	12/2011	Wenchell	N/A	N/A
8114123	12/2011	Brenzel	N/A	N/A

8118207	12/2011	Racenet	N/A	N/A
8123101	12/2011	Racenet	N/A	N/A
8128642	12/2011	Heeps	N/A	N/A
8152041	12/2011	Kostrzewski	N/A	N/A
8152823	12/2011	Kassab	N/A	N/A
8157149	12/2011	Olson	N/A	N/A
8157152	12/2011	Holsten	N/A	N/A
8162197	12/2011	Mastri	N/A	N/A
8172122	12/2011	Kasvikis	N/A	N/A
8181837	12/2011	Roy	N/A	N/A
8186558	12/2011	Sapienza	N/A	N/A
8187286	12/2011	Jugenheimer	N/A	N/A
8191752	12/2011	Scirica	N/A	N/A
8192460	12/2011	Orban, III	N/A	N/A
8196796	12/2011	Shelton, IV	N/A	N/A
8205620	12/2011	Taylor	N/A	N/A
8205780	12/2011	Sorrentino	N/A	N/A
8205781	12/2011	Baxter, III	N/A	N/A
8210412	12/2011	Marczyk	N/A	N/A
8210413	12/2011	Whitman	N/A	N/A
8210414	12/2011	Bettuchi	N/A	N/A
8210416	12/2011	Milliman	N/A	N/A
8215532	12/2011	Marczyk	N/A	N/A
8220690	12/2011	Hess	N/A	N/A
8221445	12/2011	Van Tassel	N/A	N/A
8225980	12/2011	Rivera	N/A	N/A
8231040	12/2011	Zemlok	N/A	N/A
8235272	12/2011	Nicholas	N/A	N/A
8235274	12/2011	Cappola	N/A	N/A
8236015	12/2011	Bettuchi	N/A	N/A
8240536	12/2011	Marczyk	N/A	N/A
8240537	12/2011	Marczyk	N/A	N/A
8241322	12/2011	Whitman	N/A	N/A
8245900	12/2011	Scirica	N/A	N/A
8245901	12/2011	Stopek	N/A	N/A
8246634	12/2011	Huitema	N/A	N/A
8256654	12/2011	Bettuchi	N/A	N/A
8256655	12/2011	Sniffin	N/A	N/A
8256656	12/2011	Milliman	N/A	N/A
8267849	12/2011	Wazer	N/A	N/A
8272552	12/2011	Holsten	N/A	N/A
8272553	12/2011	Mastri	N/A	N/A
8272554	12/2011	Whitman	N/A	N/A
8276594	12/2011	Shah	N/A	N/A
8276800	12/2011	Bettuchi	N/A	N/A
8276801	12/2011	Zemlok	N/A	N/A
8276802	12/2011	Kostrzewski	N/A	N/A
8281973	12/2011	Wenchell	N/A	N/A
8281974	12/2011	Hessler	N/A	N/A
8281975	12/2011	Criscuolo	N/A	N/A

8286847	12/2011	Taylor	N/A	N/A
8286848	12/2011	Wenchell	N/A	N/A
8287563	12/2011	Khairkhahan	N/A	N/A
8292146	12/2011	Holsten	N/A	N/A
8292147	12/2011	Viola	N/A	N/A
8292148	12/2011	Viola	N/A	N/A
8292149	12/2011	Ivanko	N/A	N/A
8292150	12/2011	Bryant	N/A	N/A
8292151	12/2011	Viola	N/A	N/A
8292152	12/2011	Milliman	N/A	N/A
8292153	12/2011	Jankowski	N/A	N/A
8292154	12/2011	Marczyk	N/A	N/A
8292155	12/2011	Shelton, IV	N/A	N/A
8292156	12/2011	Kostrzewski	N/A	N/A
8292158	12/2011	Sapienza	N/A	N/A
8403197	12/2012	Viola	N/A	N/A
8561872	12/2012	Wheeler	N/A	N/A
8647350	12/2013	Mohan	N/A	N/A
8685055	12/2013	Vantassel	N/A	N/A
9072536	12/2014	Shelton, IV	N/A	N/A
9232941	12/2015	Mandakolathur Vasudevan	N/A	N/A
9289211	12/2015	Williams	N/A	N/A
9375218	12/2015	Wheeler	N/A	N/A
9861359	12/2017	Shelton, IV	N/A	N/A
10595861	12/2019	Wheeler	N/A	N/A
2001/0005787	12/2000	Oz	N/A	N/A
2001/0014811	12/2000	Hussein	N/A	N/A
2001/0016748	12/2000	Tanner	N/A	N/A
2001/0034536	12/2000	Looper	N/A	N/A
2001/0039423	12/2000	Skiba	N/A	N/A
2001/0041914	12/2000	Frazier	N/A	N/A
2002/0022860	12/2001	Borillo	N/A	N/A
2002/0022880	12/2001	Melvin	N/A	N/A
2002/0032454	12/2001	Durgin	N/A	N/A
2002/0047035	12/2001	Coleman	N/A	N/A
2002/0049457	12/2001	Kaplan	N/A	N/A
2002/0055750	12/2001	Durgin	N/A	N/A
2002/0065535	12/2001	Kneifel	N/A	N/A
2002/0068945	12/2001	Robert	N/A	N/A
2002/0072759	12/2001	Fry	N/A	N/A
2002/0082621	12/2001	Schurr	N/A	N/A
2002/0103492	12/2001	Kaplan	N/A	N/A
2002/0111637	12/2001	Kaplan	N/A	N/A
2002/0120281	12/2001	Overaker	N/A	N/A
2002/0138086	12/2001	Robert	N/A	N/A
2002/0151889	12/2001	Swanson	N/A	N/A
2002/0173848	12/2001	Sachs	N/A	N/A
2002/0177863	12/2001	Mandel	N/A	N/A
2002/0183770	12/2001	Anderson	N/A	N/A

2002/0183771	12/2001	Burbank	N/A	N/A
2002/0183785	12/2001	Howell	N/A	N/A
2002/0198549	12/2001	Robert	N/A	N/A
2003/0023266	12/2002	Borillo	N/A	N/A
2003/0065340	12/2002	Geitz	N/A	N/A
2003/0078603	12/2002	Schaller	N/A	N/A
2003/0114865	12/2002	Sater	N/A	N/A
2003/0120337	12/2002	Van Tassel	N/A	N/A
2003/0125729	12/2002	Hooven	N/A	N/A
2003/0125755	12/2002	Schaller	N/A	N/A
2003/0130670	12/2002	Anderson	N/A	N/A
2003/0153930	12/2002	De Canniere	N/A	N/A
2003/0158464	12/2002	Bertolero	N/A	N/A
2003/0158604	12/2002	Cauthen	N/A	N/A
2003/0181929	12/2002	Geitz	N/A	N/A
2003/0191479	12/2002	Thornton	N/A	N/A
2003/0191494	12/2002	Gray	N/A	N/A
2003/0191526	12/2002	Van Tassel	N/A	N/A
2003/0195531	12/2002	Gardiner	N/A	N/A
2003/0216757	12/2002	Gerberding	N/A	N/A
2003/0220660	12/2002	Kortenbach	N/A	N/A
2003/0220667	12/2002	van der Burg	N/A	N/A
2003/0225421	12/2002	Peavey	N/A	N/A
2003/0225423	12/2002	Huitema	N/A	N/A
2003/0225424	12/2002	Benderev	N/A	N/A
2003/0229367	12/2002	Viola	N/A	N/A
2003/0233095	12/2002	Urbanski	N/A	N/A
2003/0236537	12/2002	Hart	N/A	N/A
2004/0006353	12/2003	Bosley	N/A	N/A
2004/0024414	12/2003	Downing	N/A	N/A
2004/0030335	12/2003	Zenati	N/A	N/A
2004/0034375	12/2003	Ruiz	N/A	N/A
2004/0049210	12/2003	Vantassel	N/A	N/A
2004/0059354	12/2003	Smith	N/A	N/A
2004/0059359	12/2003	Wilson	N/A	N/A
2004/0073234	12/2003	Chu	N/A	N/A
2004/0073241	12/2003	Barry	N/A	N/A
2004/0089312	12/2003	Jordan	N/A	N/A
2004/0092973	12/2003	Chanduszko	N/A	N/A
2004/0093024	12/2003	Lousararian	N/A	N/A
2004/0097980	12/2003	Ferree	N/A	N/A
2004/0097982	12/2003	Jugenheimer	N/A	N/A
2004/0111100	12/2003	Benderev	N/A	N/A
2004/0116948	12/2003	Robert	N/A	N/A
2004/0122456	12/2003	Saadat	N/A	N/A
2004/0122467	12/2003	Vantassel	N/A	N/A
2004/0127919	12/2003	Hugh	N/A	N/A
2004/0127935	12/2003	Vantassel	N/A	N/A
2004/0133214	12/2003	Kayan	N/A	N/A
2004/0133215	12/2003	Baxter	N/A	N/A

2004/0133219	12/2003	Forsell	N/A	N/A
2004/0138683	12/2003	Shelton	N/A	N/A
2004/0186486	12/2003	Roue	N/A	N/A
2004/0193188	12/2003	Francesce	N/A	N/A
2004/0199178	12/2003	Small	N/A	N/A
2004/0230222	12/2003	Van Der Burg	N/A	N/A
2004/0260314	12/2003	Lizardi	N/A	N/A
2005/0004652	12/2004	Van Der Burg	N/A	N/A
2005/0021061	12/2004	Dennis	N/A	N/A
2005/0021062	12/2004	Dennis	N/A	N/A
2005/0027308	12/2004	Davis	N/A	N/A
2005/0049573	12/2004	Van Tassel	N/A	N/A
2005/0059988	12/2004	Danitz	N/A	N/A
2005/0125010	12/2004	Smith	N/A	N/A
2005/0139635	12/2004	Wukusick	N/A	N/A
2005/0143767	12/2004	Kimura	N/A	N/A
2005/0146069	12/2004	Kanan	N/A	N/A
2005/0149068	12/2004	Williams	N/A	N/A
2005/0149069	12/2004	Bertolero	N/A	N/A
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2005/0149989	12/2004	Lupoi	N/A	N/A
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2005/0165421	12/2004	Wilson	N/A	N/A
2005/0165422	12/2004	Wilson	N/A	N/A
2005/0165423	12/2004	Gallagher	N/A	N/A
2005/0165424	12/2004	Gallagher	N/A	N/A
2005/0171560	12/2004	Hughett	N/A	N/A
2005/0177180	12/2004	Kaganov	N/A	N/A
2005/0177182	12/2004	Van Der Burg	N/A	N/A
2005/0177224	12/2004	Fogarty	N/A	N/A
2005/0177232	12/2004	Ashton	N/A	N/A
2005/0187569	12/2004	Dahl	N/A	N/A
2005/0234543	12/2004	Glaser	N/A	N/A
2005/0251184	12/2004	Anderson	N/A	N/A
2005/0273122	12/2004	Theroux	N/A	N/A
2005/0277955	12/2004	Palmer	N/A	N/A
2005/0277959	12/2004	Cosgrove	N/A	N/A
2006/0004388	12/2005	Whayne	N/A	N/A
2006/0004390	12/2005	Rosenberg	N/A	N/A
2006/0020162	12/2005	Whayne	N/A	N/A
2006/0020271	12/2005	Stewart	N/A	N/A
2006/0020275	12/2005	Goldfarb	N/A	N/A
2006/0025812	12/2005	Shelton	N/A	N/A
2006/0100644	12/2005	Mola	N/A	N/A
2006/0100649	12/2005	Hart	N/A	N/A
2006/0111735	12/2005	Crainich	N/A	N/A
2006/0151567	12/2005	Roy	N/A	N/A
2006/0217749	12/2005	Wilson	N/A	N/A
2006/0264979	12/2005	Shepard	N/A	N/A
2006/0264987	12/2005	Sgro	N/A	N/A

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2007/0016228	12/2006	Salas	N/A	N/A
2007/0060951	12/2006	Shannon	N/A	N/A
2007/0073274	12/2006	Chin	N/A	N/A
2007/0083218	12/2006	Steven	N/A	N/A
2007/0083227	12/2006	Van Der Burg	N/A	N/A
2007/0106314	12/2006	Dunn	N/A	N/A
2007/0112365	12/2006	Hilal	N/A	N/A
2007/0118161	12/2006	Kennedy	N/A	N/A
2007/0118163	12/2006	Boudreaux	N/A	N/A
2007/0135826	12/2006	Zaver	N/A	N/A
2007/0149988	12/2006	Michler	N/A	N/A
2007/0149989	12/2006	Santilli	N/A	N/A
2007/0149995	12/2006	Quinn	N/A	N/A
2007/0162060	12/2006	Wild	N/A	N/A
2007/0167964	12/2006	Willis	N/A	N/A
2007/0179512	12/2006	Olsen	N/A	N/A
2007/0185506	12/2006	Jackson	N/A	N/A
2007/0191868	12/2006	Theroux	N/A	N/A
2007/0203391	12/2006	Bloom	N/A	N/A
2007/0208357	12/2006	Houser	N/A	N/A
2007/0213747	12/2006	Monassevitch	N/A	N/A
2007/0246505	12/2006	Pace-Florida	N/A	N/A
2007/0250086	12/2006	Wiley	N/A	N/A
2007/0260278	12/2006	Wheeler	N/A	N/A
2007/0265640	12/2006	Kortenbach	N/A	N/A
2007/0265641	12/2006	Roue	N/A	N/A
2007/0265642	12/2006	Chanduszko	N/A	N/A
2007/0270891	12/2006	McGuckin	N/A	N/A
2007/0276417	12/2006	Mendes, Jr.	N/A	N/A
2007/0299469	12/2006	Carpenter	N/A	N/A
2008/0009866	12/2007	Alamin	N/A	N/A
2008/0021466	12/2007	Shadduck	N/A	N/A
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2008/0027471	12/2007	Hauri	N/A	N/A
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2008/0033241	12/2007	Peh	N/A	N/A
2008/0033457	12/2007	Francischelli	N/A	N/A
2008/0039879	12/2007	Chin	N/A	N/A
2008/0039922	12/2007	Miles	N/A	N/A
2008/0060658	12/2007	Doorschodt	N/A	N/A
2008/0071294	12/2007	Bender	N/A	N/A
2008/0078800	12/2007	Hess	N/A	N/A
2008/0078802	12/2007	Hess	N/A	N/A
2008/0078803	12/2007	Shelton	N/A	N/A
2008/0078804	12/2007	Shelton	N/A	N/A
2008/0078806	12/2007	Omaits	N/A	N/A
2008/0078808	12/2007	Hess	N/A	N/A
2008/0097571	12/2007	Denison	N/A	N/A
2008/0105265	12/2007	Pannell	N/A	N/A

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2008/0132891	12/2007	Nobis	N/A	N/A
2008/0140095	12/2007	Smith	N/A	N/A
2008/0147083	12/2007	Vold	N/A	N/A
2008/0177292	12/2007	Jacobs	N/A	N/A
2008/0183283	12/2007	Downing	N/A	N/A
2008/0208324	12/2007	Glithero	N/A	N/A
2008/0215090	12/2007	Gonzales	N/A	N/A
2008/0234785	12/2007	Nakayama	N/A	N/A
2008/0249547	12/2007	Dunn	N/A	N/A
2008/0269787	12/2007	Laufer	N/A	N/A
2008/0287989	12/2007	Weisel	N/A	N/A
2008/0294179	12/2007	Balbierz	N/A	N/A
2008/0300686	12/2007	Khoo	N/A	N/A
2008/0312670	12/2007	Lutze	N/A	N/A
2008/0314956	12/2007	Boudreaux	N/A	N/A
2008/0314961	12/2007	Boudreaux	N/A	N/A
2008/0319456	12/2007	Hart	N/A	N/A
2009/0001121	12/2008	Hess	N/A	N/A
2009/0001124	12/2008	Hess	N/A	N/A
2009/0005808	12/2008	Hess	N/A	N/A
2009/0012545	12/2008	Williamson, IV	N/A	N/A
2009/0020584	12/2008	Soltz	N/A	N/A
2009/0048665	12/2008	Miron	N/A	N/A
2009/0054916	12/2008	Meier	N/A	N/A
2009/0088783	12/2008	Kennedy	N/A	N/A
2009/0093826	12/2008	Warder-Gabaldon	N/A	N/A
2009/0105731	12/2008	Priewe	N/A	N/A
2009/0112249	12/2008	Miles	N/A	N/A
2009/0118748	12/2008	Pugsley	N/A	N/A
2009/0131959	12/2008	Rolland	N/A	N/A
2009/0138028	12/2008	Wells	N/A	N/A
2009/0163937	12/2008	Kassab	N/A	N/A
2009/0171380	12/2008	Whiting	N/A	N/A
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2009/0209986	12/2008	Stewart	N/A	N/A
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2009/0240267	12/2008	Crawley	N/A	N/A
2009/0240268	12/2008	Kassab	N/A	N/A
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2010/0125288	12/2009	Gelfand	N/A	N/A
2010/0130965	12/2009	Sibbitt, Jr.	N/A	N/A
2010/0137885	12/2009	Ortiz	N/A	N/A
2010/0145361	12/2009	Francischelli	N/A	N/A
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2010/0185219	12/2009	Gertzman	N/A	N/A
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2010/0191279	12/2009	Kassab	N/A	N/A
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2010/0217314	12/2009	Holsten	N/A	N/A
2010/0222789	12/2009	Gelbart	N/A	N/A
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2010/0228279	12/2009	Miles	N/A	N/A
2010/0228285	12/2009	Miles	N/A	N/A
2010/0234862	12/2009	Patel	N/A	N/A
2010/0241139	12/2009	Harshman	N/A	N/A
2010/0256660	12/2009	Anderson	N/A	N/A
2010/0264194	12/2009	Huang	227/180.1	A61B 17/07207
2010/0286718	12/2009	Kassab	N/A	N/A
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2010/0292719	12/2009	Ducharme	N/A	N/A
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2010/0324585	12/2009	Miles	N/A	N/A
2010/0324586	12/2009	Miles	N/A	N/A
2010/0324587	12/2009	Miles	N/A	N/A
2010/0324588	12/2009	Miles	N/A	N/A
2010/0331862	12/2009	Monassevitch	N/A	N/A
2010/0331880	12/2009	Stopek	N/A	N/A
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Background/Summary

RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 16/599,779, filed Oct. 11, 2019, which is a divisional of U.S. patent application Ser. No. 15/037,963, filed May 19, 2016, now U.S. Pat. No. 10,485,545, which is a national phase

application of International Patent Application PCT/US2014/066438, filed Nov. 19, 2014, which claims the benefit of priority to provisional U.S. Patent Application Ser. No. 61/906,290 filed Nov. 19, 2013. The disclosures of U.S. patent application Ser. No. 16/599,779, U.S. patent application Ser. No. 15/037,963, International Patent Application PCT/US2014/066438, and provisional U.S. Patent Application Ser. No. 61/906,290 are hereby incorporated by reference in their entireties.

FIELD OF THE DISCLOSURE

(1) The present disclosure pertains broadly to the field of fasteners and/or applicators. More specifically, the disclosure relates to surgical applicators of implants and/or fasteners, including but not limited to sterilized fasteners such as staples.

BACKGROUND

(2) Atrial fibrillation is a relatively common condition characterized by a very rapid heartbeat of the left and right atrium. While atrial fibrillation is not normally fatal itself, it has been associated with an increased risk of stroke. It is believed that the rapid heartbeat causes blood to pool in the left atrial appendage which causes emboli that are released into the left atrium from where they can enter the cerebral vasculature, thus causing a stroke. In addition to stroke, the emboli can enter coronary circulation, potentially causing myocardial infarction, or can enter peripheral circulation, potentially causing peripheral vascular disease.

(3) The risk of stroke in patients suffering from atrial fibrillation can be reduced in a variety of ways. For example, blood thinning drugs can be used to reduce the risk of clot formation. The use of blood thinners, however, is contraindicated in patients at risk of bleeding disorders. More aggressive treatment protocols have been proposed which involve closing the left atrial appendage. Closure and excision may be performed in open surgical procedures, typically requiring the patient to be placed on bypass and the chest to be opened through the sternum. Alternatively, thoracoscopic and other less invasive procedures have been proposed. U.S. Pat. No. 5,306,234 teaches the performance of heating heart procedures using otherwise conventional surgical techniques. The use of conventional techniques through small chest penetrations while the heart is beating can be difficult to perform. U.S. Pat. No. 5,865,791 describes an intravascular approach where tools are introduced through the vasculature and passed into the left atrium. The tools are used to ablate or fuse the left atrial appendage from the inside using energy, adhesives, or the like. The '791 patent also describes a thoracoscopic procedure where a tether is placed over the neck of the atrial appendage and tied off to achieve isolation. The '791 patent still further suggests other closure elements including sutures, staples, shape-memory wires, biocompatible adhesives, and the like. U.S. Pat. No. 6,488,689 describes a transpericardial procedure where the distal tip of the left atrial appendage is grasped and pulled backwardly through a capture loop which encircles the base of the left atrial appendage.

(4) A compliant closure structure for the scaling bodily structures such as the left atrial appendage is described in co-pending, commonly owned U.S. Patent Publication 2007/17260278 (application Ser. No. 11/744,135), the full disclosure of which is incorporated herein by reference in its entirety. The compliant structure described in the '278 publication comprises an elastomeric body having a pair of opposed legs which may be arranged in an oval or a U-shaped configuration to define an opening therebetween. By placing the opening between the legs over the left atrial appendage and aligning it with the base of the appendage, the structure may be closed to provide the desired sealing. To hold the structure closed, a number of discrete, axially spaced-apart tissue penetrating fasteners are arranged along the lengths of each of the legs. By compressing the legs together to press-fit the closure devices, the compliant structure may be closed to provide a compliant seal which effectively isolates the left atrial appendage.

(5) The '278 publication describes a particular delivery tool for the compliant closure structure. The delivery tool includes jaws which can be inserted into the legs of the closure structure and actuated to close the jaws in the legs over the left atrial appendage. The jaws further include comb studs

which engage and press fit the closure devices in order to hold the compliant structure in its closed, sealing configuration. The studs are intended to be retracted to allow the delivery tool to be removed.

(6) Although functional, the delivery tool of the '278 publication has certain shortcomings. For example, the actuation of the jaws and retraction of the comb studs can be performed out of order, increasing the risk that the delivery of the compliant structure will fail. Moreover, positioning and orientation of the delivery tool can be difficult, particularly when the tool is introduced through an intercostal penetration to access the left atrium. Additionally, the jaws in the device of the '278 publication are attached in the axial plane of the device shaft. Such a straight line of attachment can make it more difficult to align the jaws with the base of the appendage and across the os (i.e., ostium) of the atrium leading into the appendage. If the closure device is not aligned across the base to completely close the os, gaps or openings (referred to as “cul-de-sacs”) can remain at the site of closure, increasing the risk of thrombus formation in the atrium. The importance of forming a complete seal of the os which is free from such cul-de-sacs is discussed in Sulzberg et al. (2008) Eur. J. Cardiothoracic Surg. 34:766-770.

(7) For these reasons, it would be desirable to provide improved delivery tools for use with the tissue closure devices described in U.S. Patent Publication 2007/0260278. It would be further desirable if the delivery tools and methods of their use were compatible with the delivery of other tissue closure devices and for procedures in addition to closure of the left atrial appendage.

SUMMARY

(8) A surgical applicator for a fastener according to an example embodiment comprises a handle, a first trigger coupled movably with respect to the handle, and a second trigger coupled movably with respect to the first trigger, but operatively arranged to move in tandem with the first trigger during movement of the first trigger. An interlock is coupled to the second trigger and operatively arranged to selectively prevent movement of the second trigger relative to the first trigger depending on a position of the first trigger with respect to the handle.

(9) According to an example embodiment, the interlock permits movement of the second trigger relative to the first trigger when the first trigger is moved from a first position to a second position, and restricts movement of the second trigger relative to the first trigger when the first trigger is not in the second position.

(10) According to an example embodiment, a second interlock is coupled to the first trigger and operatively arranged to selectively prevent movement of the first trigger relative to the handle depending on a position of second trigger.

(11) According to an example embodiment, a jaw assembly is selectively opened and closed via movement of the first trigger.

(12) According to an example embodiment, movement of the second trigger selectively retracts one or more fastener supporting structures of the jaw assembly with respect to at least one jaw of the jaw assembly.

(13) According to an example embodiment, the interlock includes a cam follower engaged with a cam channel.

(14) According to an example embodiment, the handle includes the cam channel and the second trigger includes the cam follower.

(15) According to an example embodiment, the cam channel has a first leg that permits tandem movement of the second trigger with the first trigger during closing of the first trigger and a shoulder that prevents movement of the second trigger relative to the first trigger until the first trigger is closed.

(16) According to an example embodiment, the cam channel has a second leg that permits tandem movement of the second trigger with the first trigger during re-opening of the first trigger after the second trigger has been closed relative to the first trigger.

(17) According to an example embodiment, the cam channel includes at least one leg formed

concentrically with respect to a first pivot about which the first trigger is rotatably coupled to the handle and a transverse portion funned concentrically with respect to a second pivot about which the second trigger is rotatably coupled to the first trigger.

(18) According to an example embodiment, the at least one leg includes two legs, and the transverse portion is connected between the two legs.

(19) A fastener applicator according to an example embodiment comprises a handle, a first trigger coupled rotatably with respect to the handle, and a second trigger coupled rotatably with respect to the handle. An interlock is disposed with the second trigger and comprises a cam follower engaged in a cam channel that selectively prevents rotation of the second trigger relative to the first trigger depending on a location of the cam follower within the cam channel. The location of the cam follower in the cam channel is set by a position of the first trigger.

(20) According to an example embodiment, the handle includes the cam channel and the second trigger includes the cam follower.

(21) According to an example embodiment, the cam channel has at first leg that permits tandem movement of the second trigger with the first trigger during closing of the first trigger and a shoulder that prevents movement of the second trigger relative to the first trigger until the first trigger is closed.

(22) According to an example embodiment, the cam channel has a second leg that permits tandem movement of the second trigger with the first trigger during re-opening of the first trigger after the so and trigger has been closed relative to the first trigger.

(23) A fastener applicator according to an example embodiment comprises a handle, a shaft extending distally from the handle and a jaw assembly at a distal end of the shaft having one or more fastener supporting structures. A first trigger is coupled movably with respect to the handle and to the jaw assembly such that movement of the first trigger relative to the handle selectively closes the jaw assembly. A second trigger is coupled movably with respect to the first trigger, the second trigger coupled to the jaw assembly such that movement of the second trigger relative to the first trigger selectively retracts die one or more fastener supporting structures; and an interlock coupled to the second trigger and operatively arranged to selectively prevent movement of the second trigger relative to the first trigger depending on a position of the first trigger.

(24) According to an example embodiment, the interlock permits movement of the second trigger relative to the first trigger when the first trigger is closed and prevents movement of the second trigger relative to the first trigger when the first trigger is open.

(25) According to an example embodiment, a second interlock is coupled to the first trigger and operatively arranged to selectively prevent movement of the first trigger relative to the handle depending on a position of second trigger.

(26) According to an example embodiment, the interlock includes a cam follower engaged with a cam channel.

(27) According to an example embodiment, the handle includes the cam channel and the second trigger includes the cam follower.

(28) According to an example embodiment, the second trigger is movable in tandem with the first trigger during movement of the first trigger.

(29) A fastener applicator according to an example embodiment comprises a handle, a first trigger movable with respect to the handle between a first position and a second position, and a second trigger movable with respect to the first trigger between an initial position and an actuated position. A first interlock is coupled to the second trigger and operatively arranged to selectively prevent movement of the second trigger relative to the first trigger until the first trigger is moved into the second position. A second interlock is coupled to the first trigger and operatively unarmed to selectively prevent movement of the first trigger relative to the handle until the second trigger is moved into the actuated position.

(30) According to an example embodiment, a pair of jaws is included, with the first trigger coupled

to the pair of jaws for setting the pair of jaws in an open configuration when the first trigger is in the first position and in a closed configuration when the first trigger is in the second position.

(31) According to an example embodiment, one or more fastener supporting structures are disposed with the pair of jaws. The second trigger is coupled to the one or more fastener supporting structures for setting the one or more fastener supporting structures in a deployed configuration when the second trigger is in the initial position and in a retracted configuration when the second trigger is in the actuated position.

(32) A method of operating a fastener applicator according to an example embodiment comprises:

(i) repositioning a first trigger of the fastener applicator with respect to a handle from a first position to a second position, (ii) moving a second trigger in tandem with the first trigger, (iii) preventing repositioning of a second trigger between an initial position and an actuated position relative to the first trigger with an interlock coupled to the second trigger until the first trigger is moved to the second position, and (iv) repositioning the second trigger to the actuated position relative to the first trigger after the first trigger is moved to the second position.

(33) A method of operating a fastener applicator according to an example embodiment comprises:

(i) moving a first trigger of the fastener applicator with respect to a handle from a first position to a second position, (ii) preventing movement of a second trigger between an initial position and an actuated position relative to the first trigger with an interlock coupled to the second trigger until the first trigger is moved to the second position, (iii) rearranging a cam follower and a cam channel of the interlock with respect to each other due to the moving of the first trigger to the second position, and (iv) moving the second trigger to the actuated position relative to the first trigger after the first trigger is moved to the second position.

(34) A method of operating a fastener applicator according to an example embodiment comprises:

(i) moving a first trigger of the fastener applicator with respect to a handle from a first position to a second position, (ii) preventing movement of a second trigger between an initial position and an actuated position relative to the first trigger with a first interlock coupled to the second trigger until the first trigger is moved to the second position, (iii) moving the second trigger to the actuated position relative to the first trigger after the first trigger is moved to the second position, (iv) preventing movement of the first trigger relative to the handle back to the first position with a second interlock coupled to the first trigger until the second trigger is moved to the actuated position, and (v) moving the first trigger back to the first position relative to the handle after the second trigger is moved to the actuated position.

(35) A method of operating a fastener applicator according to an example embodiment comprises:

(i) repositioning a first trigger of the fastener applicator with respect to a handle from a first position to a second position, (ii) closing a jaw assembly coupled to the handle of the fastener applicator due to movement of the first trigger to the second position, (iii) preventing repositioning of a second trigger between an initial position and an actuated position relative to the first trigger with an interlock coupled to the second trigger until the first trigger is moved to the second position, (iv) repositioning the second trigger to the actuated position relative to the first trigger after the first trigger is moved to the second position, and (v) retracting one or more fastener supporting structures comprising at least one protrusion movably extendable and retractable with respect to at least one jaw of the jaw assembly of the fastener applicator due to repositioning of the second trigger to the actuated position.

(36) According to an example embodiment, the fastener applicator comprises a second interlock coupled to the first trigger and the method further comprises preventing movement of the first trigger from the second position to the first position with the second interlock until the second trigger is moved to the actuated position relative to the first trigger, and repositioning the first trigger from the second position back to the first position after the second trigger is moved to the actuated position relative to the first trigger.

(37) According to an example embodiment, the interlock includes a cam follower engaged with a

cam channel.

(38) According to an example embodiment, rotating the first trigger from the first position to the second position includes traversing the cam follower along a first leg of the channel.

(39) According to an example embodiment, the first leg is formed concentrically with respect to a pivot about which the first trigger rotates.

(40) According to an example embodiment, rotating the second trigger to the actuated position relative to the first trigger includes traversing the cam follower along a portion of the channel transverse to the first leg.

(41) According to an example embodiment, the handle includes the cam channel and the second trigger includes the cam follower.

(42) According to an example embodiment, rotating the first trigger from the first position to the second position including moving the second trigger in tandem with the first trigger.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The following descriptions should not be considered limiting in any way. With reference to the accompanying drawings, like elements are numbered alike:

(2) FIG. 1 is a perspective view of a closure device applicator constructed in accordance with to an example embodiment of the present disclosure;

(3) FIG. 2 is a partial cross-sectional elevation view of the closure device applicator of FIG. 1, shown with the jaws open and rotated 90° relative to the position shown in FIG. 1;

(4) FIG. 3 is a partial, cross-sectional elevation view of the device applicator similar to FIG. 2, shown with the first trigger closed relative to the handle in order to close the jaws, and the second trigger open (or spaced apart) with respect to the first trigger;

(5) FIG. 4 is a partial, cross-sectional elevation view of the closure device applicator similar to that shown in FIGS. 2 and 3 with the second trigger closed relative to the first trigger demonstrating a retraction of the closure device-engaging studs;

(6) FIG. 5 is a partial, cross-sectional elevation view of the closure device applicator of FIGS. 2-4, shown with the first trigger re-opened with respect to the handle in order to open the jaws while the closure device-engaging studs remain retracted;

(7) FIG. 6A-6D illustrate the movement of the jaw and stud engagement with the closure devices of a tissue closure device according to an example embodiment of the present disclosure;

(8) FIG. 7 illustrates an exemplary mechanism by which the closure device-engaging studs are retracted within the jaws according to an example embodiment of the present disclosure;

(9) FIG. 8 illustrates an interlock formed between a cam channel, slot, or groove in a handle of the applicator of FIG. 1, shown in cross-section, and a cam follower projecting from a second trigger of the applicator of FIG. 1, shown in a perspective and exploded view orientation according to an embodiment of the present disclosure;

(10) FIGS. 9A-9E illustrate various positions of the cam follower of FIG. 4 along the cam channel of the interlock as first and second triggers of the applicator are moved through the configurations depicted in FIGS. 2-5;

(11) FIG. 10 illustrates operation of the interlock of FIG. 8 with respect to the first and second triggers of the applicator of FIG. 1; and

(12) FIG. 11 illustrates use of the applicator according to an example embodiment of the present disclosure, wherein the applicator is used for closing a closure device over a left atrial appendage.

DETAILED DESCRIPTION

(13) A detailed description of one or more embodiments of the disclosed apparatus and method are presented herein by way of exemplification and not limitation with reference to the Figures.

(14) Embodiments of the present disclosure provide alternative and improved apparatuses, systems, and methods for deploying one or more closure devices or fasteners to tissue. An exemplary tissue structure of a patient at risk of stroke or other adverse events resulting from emboli released into circulation from the left atrial appendage, may include the left atrial appendage. Patients benefiting from the exemplary procedures disclosed herein will at least partially include those suffering from atrial fibrillation which can cause clot and thrombus formation in the left atrial appendage, thus increasing the chance of emboli release.

(15) Exemplary embodiments of the present disclosure provide a mechanical closure device applicator for introducing the fastener over the tissue structure, which fastener is left in place in order to close and/or seal the tissue structure. A portion of the tissue structure extending beyond the fastener may then be cut, excised, or otherwise removed, although this may be left to the physician's preference. In one embodiment, the tissue fastener comprises a compression body having at least two opposed, compliant tissue-engaging surfaces which are placed over opposite sides of the tissue structure. In this embodiment, the tissue-engaging surfaces are held together by a plurality of axially spaced-apart tissue-penetrating fasteners, which extend from one of the surfaces, through the intermediate tissue, and into the other surface to both hold the compression body in place and to apply a desired level of compression force, which is determined by both the softness of the compression body and the distance between the surfaces when they are fully attached. A well may be provided in the compression body around the tissue-penetrating barb of the fastener such that a gasket seal is formed by the compression body around the puncture site in the tissue. A stabilizing lip may be provided in one leg of the compression body to prevent a rolling motion of one leg with respect to another leg of the compression body in order to keep the two opposing soft members linearly aligned. More detailed descriptions of fasteners suitable for use with embodiments of the present disclosure are found in patent publication U.S. 2007/0260278, the full disclosure of which has been previously incorporated herein by reference.

(16) Referring now to FIG. 1, a fastener applicator **10** constructed in accordance with various embodiments of the present disclosure comprises a shaft **12** having a handle assembly **14** at its proximal end, and a jaw assembly **16** at a distal end, when individual jaws **18** and **20** are adapted to carry a fastener **22**, as best illustrated in FIGS. 6A-6D. The fastener **22** is shown in an example embodiment to have a U-shaped configuration which defines a V-shaped region **24** for receiving the left atrial appendage or other tissue structure when the jaws are open as shown in FIGS. 1, 2, and 6A. A plurality of studs **26** are formed along the inner surfaces of each jaw **20** and **18** to engage tissue-penetrating fasteners which comprise penetrating components **28** and receptacle components **30**, as best shown in FIG. 6A. The handle assembly **14** will include a handle **32**, a first trigger **34**, and a second trigger **36**.

(17) Referring now to FIGS. 2 and 6A, the fastener applicator **10** is shown in its shelf or delivery configuration with the fastener **22** received over the jaw assembly **16** and the V-shaped opening **24** in the device ready to be placed over a tissue structure such as the left atrial appendage (see FIG. 10). The triggers **34** and **36** are each respectively movable between an initial or first position and an actuated or second position. As will be discussed in more detail below, the first trigger **34** is movable between its respective positions relative to the handle **32**, while the second trigger **36** is movable between its respective positions relative to the first trigger **34**. In one embodiment, in their respective first positions, the first trigger **34** and the second trigger **36** are initially in a fully open position, that is, pivoted fully away from the handle **32**. For this reason, it is to be understood that any mention of "opening" the first trigger **34** refers more generally to moving the first trigger **34** toward its first or initial position, while any mention of "closing" the first trigger **34** or the second trigger **36** refers more generally to moving the first trigger **34** or the second trigger **36** towards its respective second or actuated position.

(18) After the fastener **22** is advanced over the left atrial appendage or other target tissue structure, the jaws **18** and **20** are closed by manually pulling the first trigger **34** toward the handle **32**, as

shown in FIGS. 3 and 6B. The first trigger **34** is mounted on a pin or pivot **40**, which is shown to be fixedly secured to the handle assembly **14**, thereby rotatably coupling the first trigger **34** to the handle **32**. Closing the first trigger **34** with respect to the handle **32**, that is, pivoting about the pivot **40** from the first position of the first trigger **34** to the second position, proximally retracts the rod **42** which is linked to the first trigger **34** by a pin or pivot **41** and a lever **43**. The rod **42** is attached to a pin **44** (FIG. 6B) which is pivotally attached to the proximal ends of the jaws **18** and **20** and proximally retracts the jaws so that they are closed by the movement of slots **46** over pins **48** in an end frame **50** attached to a stationary sleeve **52**. When the jaws **18** and **20** are closed, the studs **26** engage the tissue penetrating components **28** and the tissue penetrating receptacles **30** so that they engage and lock with each other, thus closing the two legs of the fastener **22**, as shown in FIG. 6B.

(19) The second trigger **36** may be mounted on the pin **41** so that the second trigger **36** moves, actuates, or closes with respect to the handle **32** in tandem with the first trigger **34**. In this way, the second trigger **36** is rotatably coupled to the handle **32**, although indirectly via the pin **41** and the first trigger **34**. By “in tandem” it is meant that movement of the first trigger **34** translates at least partially into corresponding movement of all or portions of the second trigger **36**. In the illustrated embodiment, movement of the first trigger **34** causes a substantially equivalent amount of rotational movement of the second trigger **36**, although there could be gears, linkages, springs, or the like to cause a lesser or greater degree of movement of the second trigger **36** to result from corresponding movement of the first trigger **34**.

(20) The applicator **10** may optionally include an interlock **60** in some embodiments. According to the illustrated embodiment, as the first trigger **34** and the second trigger **36** are moved in tandem, a ratcheting tooth surface **62** of the interlock **60** is closed against pins **64**, as shown in FIG. 3, so that the first trigger **34** cannot be re-opened once the jaw assembly **16** has been closed, either completely or to a certain degree of closure or spacing between the jaws. This is advantageous as described above, since the jaws should not be completely opened prior to retracting the studs **26** by closing the second trigger **36**, as will be described below. The ratcheting tooth surface **62** can include any number of ratchets or notches such that the first trigger **34** can be progressively closed (i.e., brought into approximation of the handle **32**) by more fully driving the ratcheting tooth surface **62** along the pins **64** in the direction of closure for the first trigger **34** (e.g., from the configuration of FIG. 2 to the configuration of FIG. 3).

(21) Additionally, as the first trigger **34** is moved to its actuated, closed, or second position (i.e., moved or actuated to the configuration of FIG. 3) in order to close the jaws **18** and **20** of the jaw assembly **16**, a corner **66** (see FIG. 2) of the interlock **60** will engage an inclined surface **68** on a clicker pin **70** to disengage the clicker pin **70** from a holding pin **72**, thus allowing a spring **74** to push the pin **70** downward so that a bottom surface **76** thereof strikes the bottom of the handle **32**, thus causing a loud click to audibly alert the physician that the jaws have been closed and the interlock **60** has been engaged. This audible confirmation indicates to the physician or other medical personnel that the fastener **22** has been closed and that the studs **26** can be retracted.

(22) The studs **26** are initially in an extended configuration, as shown in FIGS. 2-3 and 6A-6B, and then retracted by closing second trigger **36** against the first trigger **34**, as shown in FIGS. 4 and 6C. Closure of the second trigger **36** relative to the first trigger **34** draws the outer sleeve **54** proximally over the stationary sleeve **52** which draws pull wires **80** proximally to full stud combs **84** or other fastener supporting structures, proximally, as shown in FIG. 7. The stud combs **84** are mounted on pins **86** which travel in slots **88** formed in each of the jaws **18** and **20**. The second trigger **36** is pivotally mounted on, or rotatably coupled to, the first trigger **34** by the pin **41** and is coupled to the stationary sleeve by pins **90** and **91** and a lever **92**. It is to be appreciated that the studs **26** can take other shapes or arrangements and/or that other fastener supporting structure (i.e., structures capable of supporting the fasteners **22** during closing and engaging thereof) can be included in lieu of the studs **26** that are either disengaged from or engaged to the fastener **22** upon intentional closure or movement of the second trigger **36**. For example, in lieu of a plurality of individual studs **26**, a

supporting structure in the form of a single unitary rail supporting all of the components **28** can be included in one embodiment.

(23) Movement of the second trigger **36** relative to the first trigger **34**, e.g., closure of the second trigger **36**, disengages the interlock **60** by engaging a ramped surface **94** attached to the interlock **60** against a fixed pin **96** in the second trigger **36**. In this way, the interlock **60** selectively permits or prevents movement, e.g., opening, of the first trigger **34** depending on a position of the second trigger **36**. In other words, movement of the second trigger **36** to its corresponding second position, relative to the first trigger **34**, releases the interlock **60**. The pin **96** follows the path shown in broken line in FIG. 3 to engage the surface **94**. The ramped surface **94** is caused to move downwardly, pulling the ratchet teeth **62** away from the fixed pins **64**, as shown in FIGS. 4 and 5. The triggers **34** and **36** are now ready to be opened in order to open the jaws, as will now be described.

(24) Referring now to FIGS. 5 and 7C, the jaws **18** and **20** may be opened by releasing manual compression on the first and second triggers **34** and **36**, allowing spring **98** which was compressed during closure of the first trigger **34** to push rod **42** (attached to pivot **44**) distally forward, opening the jaws and leaving the fastener **22** in place.

(25) FIG. 8 shows a cross-section of the handle **32** as well as the second trigger **36** exploded therefrom. An interlock **100** is included between the handle **32** and the second trigger **36** to selectively permit and prevent rotational movement of the second trigger **36** with respect to the first trigger **34**. The first trigger **34** does not form a part of the interlock **100** in the illustrated embodiment, and therefore is not illustrated in FIG. 8. However, it is to be understood that the first trigger **34** in the embodiment of FIG. 8 would be nevertheless be secured to the second trigger **36** at the pivot **41** as described herein with respect to the other Figures. Additionally, the shaft **12** and the jaw assembly **16** are not illustrated in some Figures, including FIG. 8, but can be included as described herein with respect to the other Figures.

(26) More specifically, the interlock **100** includes a cam follower **102** and a channel **104**. In the illustrated embodiment, the cam follower **102** is formed as a protrusion or projection extending from or otherwise coupled to the second trigger **36**, and is preferably round in cross section but alternatively may be of other shapes. The cam follower **102** can be integrally formed with the second trigger **36** or affixed thereto as a separate component. The channel **104** may be formed as a groove, depression, notch, slot, or the like in the handle **32**. The follower **102** is configured to engage with and/or travel along the channel **104** (and thus will be restricted to movement within the channel **104**) when the second trigger **36** is assembled with the handle **32**. It is noted that the applicator **10** can include other exemplary embodiments of a single follower **102** and a single channel **104** if desired, but for symmetry, balance, etc., the second trigger **36** may include two cam followers **102** (e.g., another follower **102** hidden from view in FIG. 8 on the opposite lateral side of the second trigger **36** and mirroring the illustrated one of the followers **102** about a plane bisecting the second trigger **36**), and the handle **32** may correspondingly include two cam grooves **104** (e.g., another channel **104** located in the half of the handle **32** not shown in FIG. 8, which mirrors the illustrated channel **104** about the plane used to form the cross-sectional view of the handle **32** in FIG. 8).

(27) The mechanical coupling of the follower **102** to the second trigger **36** prevents rotation of the second trigger **36** relative to the first trigger **34** when the follower **102** abuts against the walls or shoulders defining the channel **104**. For example, FIGS. 9A-9E show various positions of the follower **102** (without the second trigger **36** for clarity) corresponding to the various combinations of positions of the first and second triggers **34** and **36** of the applicator **10**. That is, for example, the position of the follower **102** in FIG. 9A corresponds to an initial, shelf, or delivery configuration of the applicator **10** shown in FIG. 2, i.e., before the triggers **34** or **36** have been actuated.

(28) In the position of FIG. 9A, rotation of the second trigger **36** is prevented by interference of the follower **102** with a shoulder **106** of the channel **104**. That is, movement of the second trigger **36**

relative to the first trigger **34** occurs about the pin **41**, which rotationally couples the second trigger **36** to the first trigger **34**. Thus, rotation of the second trigger **36** about the pin **41** (if unblocked) would result in the cam follower **102** following an arc defined by a circular path concentric with the pin **41**, e.g., illustrated as a circle **107** in FIG. **10**. Front FIG. **10**, it is clear that rotational movement of the cam follower **102** along an arc defined by the circle **107** is not possible due to interference between the cam follower **102** and the shoulder **106**. Accordingly, rotation of the second trigger **36** about the pin **41** relative to the first trigger **34** is prevented by the interference between the cam follower **102** and the shoulder **106**.

(29) The channel **104** is shaped with a leg **108**, which permits the second trigger **36** to travel in tandem with the first trigger **34** as the first trigger **34** is moved toward its second position, i.e., rotated about the pivot **40** toward the configuration of the applicator **10** shown in FIG. **3**. That is, again referring to FIG. **10**, tandem movement of second trigger **36** due to movement of the first trigger **34** would result in the cam follower **102** traveling along an arc defined by a circular path concentric with the pivot **40**, illustrated in FIG. **10** by a circle **109**. The first leg **108** is arranged to lie substantially along such an arc formed by the circle **109**, thereby enabling the cam follower **102** to travel along the leg **108** during rotation of the first anti second triggers **34** and **36** about the pin **40**.

(30) FIG. **4B** illustrates the position of the follower **102** in the channel **104** when the first trigger **34** is rotated partially toward the configuration of FIG. **3**, e.g., with ratcheting tooth surface **62** of the interlock **60** at most only partially driven along and engaged with the pins **64**. At the position of FIG. **9B**, the shoulder **106** continues to prevent rotation of the second trigger **36** by blocking movement of the follower **104** in the direction necessary to rotate the second trigger **36**.

(31) FIG. **9C** illustrates the position of the follower **102** after the trigger first **34** has been moved to the configuration of FIG. **3** and the ratcheting tooth surface **62** of the interlock **60** is fully driven along and engaged with the pins **64**. It can be appreciated in view of FIGS. **3** and **9C** that rotation of the second trigger **36** becomes possible at this position because the shoulder **106** no longer abuts the follower **102**. Alternatively stated, it is to be appreciated that tandem movement of the second trigger **36** with the first trigger **34** has shifted the position of the cam follower **102** and the pin **41** relative to the channel **104** (since the pin **41** and the cam follower **102** are directly coupled to the second trigger **36** in the illustrated embodiment). It is to be further understood that this shifting of the position of the pin **41** and the cam follower **102** has also moved the location of the circle **107** to align with a portion **110** of the channel **104** that is transverse to the leg **108** and arranged along an arc that is concentric with respect to the pin **41**.

(32) In view of the foregoing it can be seen that the position of the first trigger **34** selectively prevents or permits rotation of the second trigger **36**. Again, movement of the first trigger **34** causes tandem movement of the second trigger **36**, which results in the follower **102** first traversing along the leg **108** of the channel **104**. However, movement of the second trigger **36** relative to the first trigger **34** is prevented due to interference between the cam follower **102** and the shoulder **106**. Relative movement of the second trigger **36** with respect to the first trigger **34** is thus prevented until the cam follower **102** is moved to a position at which the cam follower **102** no longer abuts the shoulder **106**. FIG. **9D** shows the position of the follower **102** after movement of the second trigger **36** relative to the first trigger **34** is permitted and the second trigger **36** has also been actuated to its second position, thereby corresponding to the configuration of the applicator **10** shown in FIG. **4**.

(33) The channel **104** is shaped with another leg **112** that enables the follower **102** to continue to traverse the channel **104** such that the second trigger **36** can again move in tandem with the first trigger **34** as the first trigger **34** is moved to reopen the jaw assembly **16**, i.e., as the first trigger **34** is moved back toward its first position. It is to be appreciated that the leg **112**, similar to the leg **108**, is formed along an arc that is concentric to the pivot **40**. This accordingly enables the cam follower **102** to travel along the leg **112** as the second trigger **36** moved in tandem with the first

trigger **34** as the first trigger **34** is rotated about the pivot **40**. FIG. 9E accordingly shows the corresponding location of the follower **102** in the channel **104** when the first trigger **34** is returned to its first or initial position, which may correspond to the applicator **10** assuming the configuration of FIG. 5 in which the jaw assembly **16** is again opened.

(34) It is to be appreciated that the channel **104** can take other shapes in other embodiments depending on the desired relative movement of the first and second triggers **34** and **36** relative to each other and to the handle **32**. More specifically, the channel **104** will include ii leg or portion for each optionally desired stage of movement of the first trigger **34** and of the second trigger **36**, with the legs and/or portions of the channel **104** being concentric with respect to the pivot about which rotation is desired for each stage of movement. Thus, in order to sequentially permit (i) tandem movement of both of the first and second triggers **34** and **36** about a first pivot (e.g., the pivot **40**), then lip movement of the second trigger **36** relative to the first trigger **34** about a second pivot (e.g., the pivot **41**), and then (iii) tandem movement of the first and second triggers **34** and **36** again with respect to the first pivot, the channel **104** will include two legs (e.g., the legs **108** and **112**) that are spaced apart but both concentric with respect to the first pivot (e.g., the pivot **40**), which enables the movements of stages (i) and (iii), and a transverse portion (e.g., the portion **110**) connecting between the two legs (e.g., the legs **108** and **112**) that is formed concentrically with respect to the second pivot (e.g., the pivot **41**) to enable the movement of stage (ii).

(35) From the above description and the FIGS. 9A-9E, it is clear that the interlock **100** accordingly prevents rotation of the second trigger **36** until after the trigger first **34** is first rotated, preferably fully rotated (i.e., from a first position to a second position). Since full rotation of the first trigger **34** (e.g., as shown in FIG. 3) corresponds to the closing of the jaw assembly **16**, the interlock **100** thus advantageously selectively prevents retraction of the studs **26** (due to rotation of the second trigger **36**) until after the jaw assembly **16** is closed. Referring again to the above description, it is again noted that the interlock **60** is released depending on the position of the second trigger **36** relative to the first trigger **34**. Thus, the location of the first trigger **34** is first used to release the interlock **100**, selectively permitting movement of the second trigger **36** relative to the first trigger **34**, and thereafter the location of the second trigger **36** relative to the first trigger **34** is used to release the interlock **60**, which selectively permits movement of the first trigger **34** relative to the handle **32** back toward the initial position for the first trigger. In this way, the aforementioned order of the steps of (1) closing the jaws, (2) retracting the studs, and (3) opening the jaws is maintained and ensured by use of either or both of the interlocks **60** and **100**.

(36) It is to be appreciated that the interlock **100** can take other forms in non-illustrated embodiments. For example, in one embodiment the channel **104** is alternatively formed with the second trigger **36** and the follower **102** with the handle **32**. In another embodiment, part of the channel **104** is formed in one of the handle halves while the other is formed in the other of the handle halves.

(37) Referring now to FIG. 11, one specific example of use is illustrated. In this example, the fastener applicator **10** is used to deliver one of the fasteners **22** over the base of a left atrial appendage LAA in an open chest, beating heart procedure. The sternum S is opened, spreading the ribs R to provide a working space over the heart. After opening the pericardial sack, the heart may be lifted and turned, for example using a sheet of gauze or other material which is placed behind the heart, to expose the left atrial appendage within the opening as shown in FIG. 10. After the left atrial appendage LAA is exposed, the jaws **18** and **20** of the fastener applicator are placed around the base of the appendage by manipulating the shaft **12**, as shown in FIG. 10. The angle of the jaws **18** and **20** relative to the shaft, typically about 15°, is highly advantageous as it allows the jaws to engage the base of the left atrial appendage so that they are generally parallel with the os between the appendage and the left atrium LA. If the plane of the jaws was aligned with the shaft, it would be difficult to achieve this orientation and the risk of closing the appendage and leaving a cul-de-sac (an open space beyond the os and into interior of the left atrial appendage) is greatly increased.

Such cul-de-sac is problematic as it can be a source of clotting and, if created, must be closed in the same or later procedure. Once the jaws **18** and **20** properly position the fastener **22** about the base of the left atrial appendage, the jaws are actuated and the fastener deployed as described previously. (38) While the invention has been described with reference to an exemplary embodiment or embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the invention. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the invention without departing from the essential scope thereof. Therefore, it is intended that the invention not be limited to the particular embodiment disclosed as the best mode contemplated for carrying out this invention, but that the invention will include all embodiments falling within the scope of the claims. Also, in the drawings and the description, there have been disclosed exemplary embodiments of the invention and, although specific terms may have been employed, they are unless otherwise stated used in a generic and descriptive sense only and not for purposes of limitation, the scope of the invention therefore not being so limited. Moreover, the use of the terms first, second, etc. do not denote any order or importance, but rather the terms first, second, etc. are used to distinguish one element from another. Furthermore, the use of the terms a, an, etc. do not denote a limitation of quantity, but rather denote the presence of at least one of the referenced item.

Claims

1. A method of operating a fastener applicator, the method comprising: repositioning a first trigger of the fastener applicator with respect to a handle from a first position to a second position; closing a jaw assembly coupled to the handle of the fastener applicator due to movement of the first trigger to the second position; preventing repositioning of a second trigger between an initial position and an actuated position relative to the first trigger by an interlock coupled to the second trigger until the first trigger is moved to the second position, wherein the interlock includes a cam follower engaged with a cam channel, and wherein the handle includes the cam channel and the second trigger includes the cam follower; repositioning the second trigger to the actuated position relative to the first trigger after the first trigger is moved to the second position; and retracting one or more fastener supporting structures comprising at least one protrusion movably extendable and retractable with respect to at least one jaw of the jaw assembly of the fastener applicator due to repositioning of the second trigger to the actuated position.
 2. The method of claim 1 wherein the fastener applicator comprises a second interlock coupled to the first trigger, the method further comprising preventing movement of the first trigger from the second position to the first position with the second interlock until the second trigger is moved to the actuated position relative to the first trigger, and repositioning the first trigger from the second position back to the first position after the second trigger is moved to the actuated position relative to the first trigger.
 3. The method of claim 1, wherein rotating the first trigger from the first position to the second position includes traversing the cam follower along a first leg of the channel.
 4. The method of claim 3, wherein the first leg is formed concentrically with respect to a pivot about which the first trigger rotates.
 5. The method of claim 1, wherein rotating the second trigger to the actuated position relative to the first trigger includes traversing the cam follower along a portion of the channel transverse to the first leg.
 6. The method of claim 1, wherein rotating the first trigger from the first position to the second position includes moving the second trigger in tandem with the first trigger.
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